LiDAR-Guided Reconstruction of Axial Motion Blur with Coded Monitor Illumination

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Abstract—We built a camera system to study blur as a printed target moves toward the camera during one exposure. A fixed depth sensor records the hand push, and a monitor changes the illumination. Twelve independent pushes cover steady and coded light under two motion instructions. Our main comparison uses six coded exposures and changes only the depth history assumed in reconstruction. Measured depth improves peak signal-to-noise ratio by a mean of 3.04 decibels over a straight-line depth estimate, with a sample standard deviation of 3.67 decibels. Five of six trials improve, and the mean structural-similarity gain is 0.084. Five trials also beat the blurred input, with a different trial failing that comparison. Stationary controls, timing fits and sensitivity checks support a conditional benefit from measured motion. The experiment uses reference-assisted reconstruction, with optical timing, target pose and focus still limiting recovery.

Index Terms—Axial motion, LiDAR, coded illumination, image reconstruction, experimental characterization.

1 Introduction

A target moving toward a camera grows in the image during exposure. Different image points blur by different amounts, so a single translation-invariant convolution generally cannot describe the blur. A hand push also varies in speed. Measuring depth during exposure could give a better estimate of this changing magnification, while coded illumination changes the contribution of each target position to the photograph.

We ask whether measured depth improves reconstruction of the same coded image compared with a straight line between the same exposure-endpoint depths. The comparison holds the image, illumination history, endpoint geometry, solver and scoring region fixed. We also compare each reconstruction with the blurred input to check whether a better motion estimate actually improves the image.

We built a manual guide path and used an existing upright target stand. The camera, depth board and mounting plate came from the lab kit. Monitor illumination and concurrent image, depth and command logs let us model motion within a conventional long exposure. Our processing aligns these records, constructs a scale-blur operator and estimates the sharp image. All motion was produced by hand. This report presents the supplementary physical dataset and every primary outcome; the earlier five-trial pilot used a different protocol and is kept separate.

2 Literature Review

Raskar et al. used a fluttered shutter to encode motion during exposure and improve deblurring [1]. We borrow the idea of temporal coding, apply it through illumination and model axial scaling. Our short binary pattern was not optimized. Their lateral-motion results motivate the experiment but do not establish that coding will help this apparatus.

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We use total variation (TV) regularization to limit noise while retaining edges [2]. The solver follows an accelerated proximal-gradient scheme inspired by FISTA [3], with an approximate TV step whose numerical stability we check. Scale-invariant features support endpoint registration [4]. We evaluate pixel error with peak signal-to-noise ratio (PSNR) and local structure with structural similarity (SSIM) [5]. Including the blurred input as a baseline shows whether reconstruction improves the photograph as well as the assumed motion model.

3 Method and Experimental Apparatus

3.1 Physical implementation

The OV5647 camera, Raspberry Pi 3 B+ and LightRanger 14 Click (TMF8829) are fixed on a tripod. We retained the Lab 1 kit lens, with nominal focal length approximately 30 mm. Tape holds the depth board above the camera, leaving its optical window clear. Two team-built guide edges constrain a sliding base carrying a printed USAF target in an existing upright stand (Fig. 1). An ASUS ROG Strix XG27UCGR monitor (27-inch, native 3840×2160) illuminates the paper from the side. We checked that the selected depth zones saw the paper: placing the monitor in the ranging field had previously caused foreground returns.

The base front moves from 75 to 65 cm toward the camera. The paper lies 22 mm behind it, giving $D_{\mathrm{paper}} = D_{\mathrm{edge}} + 22\,\mathrm{mm}$. This converts the position reference; it does not measure sensor bias because the ruler zero was not surveyed against the optical origin. We kept focus, aperture, monitor position and brightness fixed. The monitor was set to 160 Hz and brightness-menu value 90; optical transition times and luminance were not measured independently.

a Manual target path and monitor



b Fixed camera and depth sensor

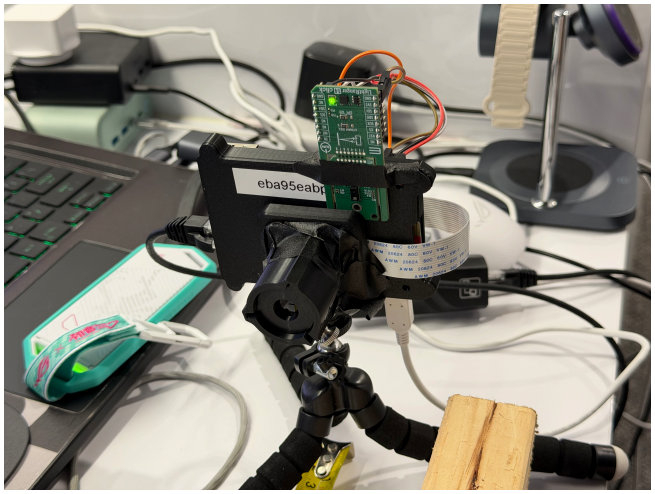


Fig. 1. Apparatus used for the supplementary experiment. (a) Printed target, manual guide path, side monitor and fixed sensors. (b) Camera and depth board on the kit plate. The team built the guide; the upright target stand and kit plate were supplied. The photographs show placement; alignment and mounting stiffness were not measured.

3.2 Acquisition and independent repeats

The dataset contains 40 runs and 1239 readable DNG images, covering calibration, stationary controls and 16 motion attempts. Twelve eligible independent pushes cover coded/steady-white illumination and smooth/variable push instructions, with three pushes in each cell. For variable pushes, the operator deliberately changed speed; for smooth pushes, the operator tried to keep the motion steady. Both instructions still allowed measured velocity to vary. We retained two pilots and two weak-motion originals outside the primary set, and collected extra independent pushes to replace the weak variable-motion attempts.

Each run saves repeated camera DNGs and continuous 8×8 depth arrays. Coded runs also save received illumination events. We select one exposure per push for the primary analysis, so multiple frames and zones do not count as independent trials. Exposure and analogue gain were requested at 1 s and 1. Metadata give an exposure of 999942 μs; actual gain was not independently checked. Acquisition is 640×480, reduced to 240×180 linear-green images for analysis.

The monitor repeats `1101001101` in nominal 200 ms states, giving a 2 s period and 60% white duty cycle. We play the code for 12 s within a 20 s recording. The camera integrates throughout each exposure while the monitor changes the paper's illumination. Since one second covers only part of the code period, we use the state history logged for that exposure to calculate its weights. White and coded runs involve separate pushes, with unmatched trajectories and light dose.

3.3 Exposure selection

We fixed an exploratory selection rule before computing reconstruction scores, without preregistration. A primary exposure needs depth coverage, reliable event coverage for coded light, an approaching target and at least 15 mm travel during integration. We reject windows crossing anomalous event intervals, then select the largest travel for smooth pushes or the largest departure from endpoint-linear depth for variable pushes. The two weak originals reach only 9.05 and 6.99 mm maximum travel and are scored separately. One coded run has an event freeze after the selected window; its earlier covered window remains usable. Event counts alone would not reveal this problem.

4 Timing and Calibration

4.1 Clock convention and measurement density

We align the records on the Pi monotonic clock. Its calendar was wrong during capture, so the original folder dates are retained only as identifiers. We take the camera interval as $[t_s - T, t_s]$, interpreting sensor timestamp t_s as the first-row readout/exposure-end time and T as the recorded exposure. Rolling-row timing was not measured. Depth and screen logs use host receipt times. The saved device ticks and player clock were not calibrated to that timebase.

Each selected coded exposure contains 10–11 valid received depth frames in one second. We take the median of the central four zones (rows 4–5, columns 4–5, one-based), requiring at least three positive zones, depth 200–2000 mm and within-region spread no greater than 50 mm. We omit the known startup sample and interpolate between received frames. Raw depth is used without subtracting the 22 mm base-to-paper offset.

4.2 Stationary checks

At paper references 672, 722 and 772 mm, three technical run summaries give depths of 664.58 ± 0.38, 715.33 ± 0.29 and 766.50 ± 0.50 mm (mean $\pm$ sample SD). The residuals are -7.42, -6.67 and -5.50 mm. One initially near-labelled record matches the far position. We retain it as a position/reference mismatch but exclude it from this summary. Because the optical origin is uncertain, these residuals cannot be assigned entirely to sensor error.

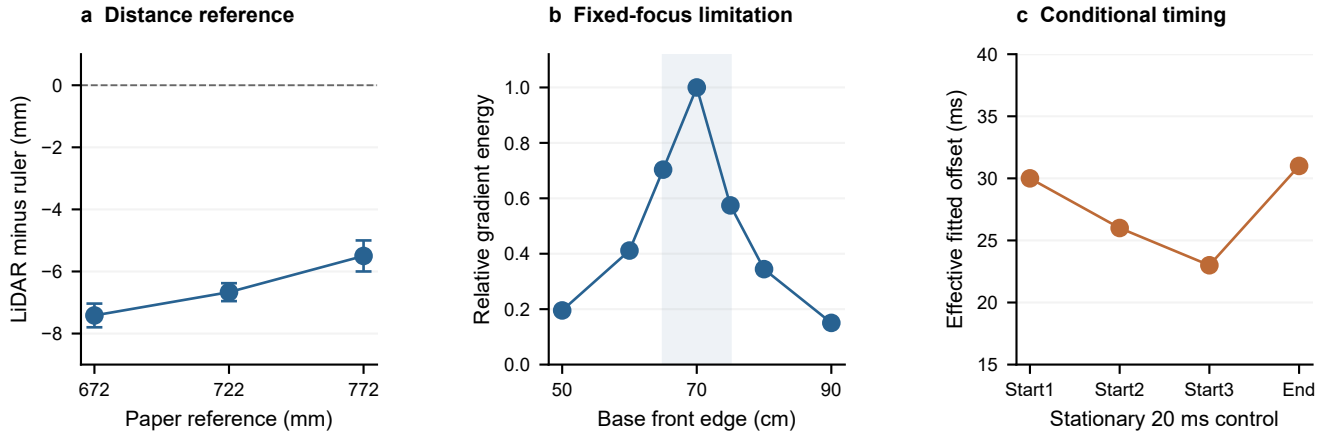


Fig. 2. Acquisition checks. (a) Depth residual against the paper reference, including the 22 mm base-to-paper offset. Bars show sample SDs of three technical run summaries; optical-origin uncertainty remains. (b) Registered gradient energy relative to 70 cm shows fixed-focus variation. It is a sharpness proxy, not a resolution measurement. Shading marks the retained 65–75 cm base range. (c) Short-exposure controls give effective fitted offsets; optical synchronization was not measured.

Stationary 19.969 ms controls give fitted screen offsets of 30, 26 and 23 ms initially, and 31 ms at the end (end-fit $R^2 = 0.985$). We use 26 ms as an effective command-to-image offset. Each one-second control has only six covered observations, with weaker transferred-fit agreement ($R^2 = 0.814$, 0.607, 0.525). The short-exposure fit therefore leaves uncertainty in the long-exposure illumination history. The fixed-patch black/white raw-intensity ratio varies from about 0.167 to 0.211; we use 0.19 in the main model.

Relative gradient energy is 0.70 at 65 cm and 0.57 at 75 cm, compared with 70 cm. It falls to 0.20 and 0.15 at 50 and 90 cm (Fig. 2). These fixed-focus checks led us to restrict motion to the narrower range, where some defocus still remains. The selected motion images and final controls are unclipped; two initial white controls each contain one clipped raw pixel. The canonical reference is the median of ten stationary 100 ms images at the nominal 70 cm base position. These frames form one reference, not ten independent experimental repeats.

5 Reconstruction and Evaluation

5.1 Linear images and empirical axial geometry

We correct DNG values using per-channel black levels and white level, average the two green samples per Bayer cell, then resize by area averaging. Inversion uses these linear images without JPEG processing or display gamma. SIFT matches and RANSAC similarity fits relate stationary early/late frames to the sharp reference; selected moving frames are excluded from endpoint fitting. Separate global 3rd/97th-percentile normalization of the input and reference assists contrast matching. All reconstructions share the input transform.

Endpoint scales calibrate a restricted axial model. With far/near depths z_f, z_n, scale ratio R, reference depth z_r and a fitted fixed pivot,

$$\beta(z) = \frac{z^{-1} - z_f^{-1}}{z_n^{-1} - z_f^{-1}}, \qquad s(z) = R^{\beta(z)-\beta(z_r)}. \tag{1}$$

This interpolates log scale over inverse depth using the measured endpoints, without requiring an exact optical-origin distance from the ruler. The main dynamic warp scales about the pivot. Endpoint rotation is used for reference registration and enters the dynamic warp only in a sensitivity check. Range alone does not recover continuous target tilt or lateral drift.

5.2 LiDAR-guided axial scale-blur model

An exposure combines differently scaled views of one sharp reference-plane image x. We write the observation y as

$$\begin{aligned} y &= A(z,u)x + \epsilon, \\ A &= \sum_{k=1}^{48} w_k W[s(z_k)], \\ w_k &= \frac{0.19 + 0.81u_k}{\sum_j (0.19 + 0.81u_j)}, \end{aligned} \tag{2}$$

where z_k is interpolated depth, $W[s(z_k)]$ scales about the fitted pivot, and u_k is the logged binary state after the timing shift. The 48 midpoint bins approximate the exposure integral from 10–11 received depth frames. Sparse bilinear interpolation implements W with replicated boundaries. The weights approximate relative irradiance, including a 0.19 floor for illumination during black states. Each column of A gives the response to an image point, a discrete point spread function (PSF). Expansion moves points farther from the pivot more strongly, so this PSF varies across the image. Reconstruction therefore inverts a spatially varying operator.

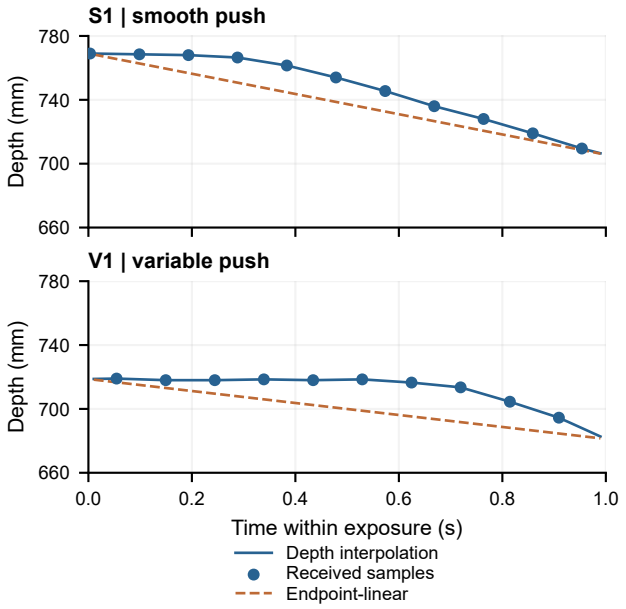


Fig. 3. Depth during two selected exposures. S1 and V1 contain 11 and 10 valid received samples. The models share the same exposure-boundary depths and differ only in the interior trajectory. A smooth-push instruction does not impose constant speed.

5.3 TV-regularised reconstruction

Once the records and endpoint geometry fix A, we estimate the image by non-blind, TV-regularised least squares:

$$\hat{x} = \arg\min_{0 \le x \le 1} \tfrac{1}{2}\|Ax - y\|_2^2 + \lambda\,\mathrm{TV}(x), \qquad \lambda = 0.002. \quad (3)$$

The data term fits the recorded blur, while TV limits unstable fluctuations. We use 320 outer iterations of an accelerated proximal-gradient scheme inspired by FISTA [3], a power-estimated step size and up to 35 inner TV iterations. The TV step is approximate, and clipping constrains intensities to $[0, 1]$. The comparator replaces measured $z(t)$ with constant depth velocity between the same boundary depths. Both methods share the image, light weights, geometry, solver and mask. We also evaluate the blurred input and denoise-only processing ($A = I$). Parameters are shared across trials.

5.4 Scoring and numerical checks

We map all 16 attempts into one canonical reference frame and fix a geometry-only intersection mask before scoring (Fig. 4). PSNR uses unit intensity range; Gaussian-window SSIM uses $\sigma = 1.5$ and population covariance [5]. Methods receive no separate post-reconstruction alignment or contrast fit. The reference is used for registration, normalization and scoring, making this a reference-assisted experiment.

Identity, row-sum and adjoint errors are below 8×10^{-7}. Synthetic inversion improves PSNR from 13.99 to 27.15 dB. Doubling 320 to 640 iterations changes PSNR by at most 0.049 dB; using 96 instead of 48 bins changes it by at most 0.207 dB. These checks test numerical consistency. The physical experiments test how well the model describes real captures.

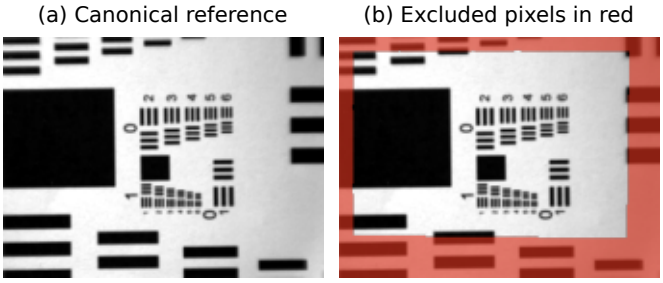


Fig. 4. Common reference and scoring mask for every attempt and method. Red marks excluded pixels. Metrics use the remaining 28327 pixels (65.6% of the 240×180 image), following the irregular interior boundary rather than a rectangular display crop.

TABLE 1
Six independent coded exposures scored on the common canonical mask. I: blurred input; C: endpoint-linear; L: LiDAR. S/V denote smooth/variable push instructions, with independent repeats numbered 1–3. Actual velocity can vary under either instruction.

	PSNR (dB)			SSIM		
Trial	I	C	L	I	C	L
S1	15.13	18.08	18.21	0.646	0.728	0.767
S2	13.57	19.15	18.04	0.564	0.789	0.765
S3	16.40	14.33	18.38	0.712	0.664	0.777
V1	18.71	14.29	19.22	0.791	0.647	0.826
V2	11.59	16.21	25.09	0.523	0.745	0.906
V3	21.45	17.70	19.07	0.878	0.803	0.840

6 Results and Analysis

6.1 Independent coded pushes

Table 1 includes every primary coded push. Targets travel 27.8–63.4 mm during the selected exposure within the nominal 100 mm path. The measured trajectories in Fig. 3 show motion that the endpoint-linear model misses.

Measured depth beats endpoint-linear in five of six trials on both metrics. Mean paired differences are $+3.038 \pm 3.666$ dB and $+0.0843 \pm 0.0797$, respectively (sample SD across six independent pushes). Figure 6 shows each paired difference and the mean with one sample SD. We report these as descriptive statistics. Against the blurred input, mean gains are $+3.523 \pm 5.421$ dB and $+0.1280 \pm 0.1485$, also with five positive trials. The median paired PSNR gain is 2.706 dB. Leaving out each push in turn gives mean gains of 1.870–3.869 dB; omitting V2, the largest gain, gives 1.870 dB. The primary result retains all six trials; these omissions only check how much each push affects the mean.

S2 loses 1.115 dB and 0.0239 SSIM to the endpoint-linear model, although both reconstructions beat its input. V3 improves over endpoint-linear but loses 2.383 dB and 0.0380 SSIM to input. V2 has the largest paired PSNR gain, 8.880 dB. Figure 5 shows V2 and both failure types. These are two different failures: measured depth can lose to the simpler motion model, and motion inversion can degrade an already better input.

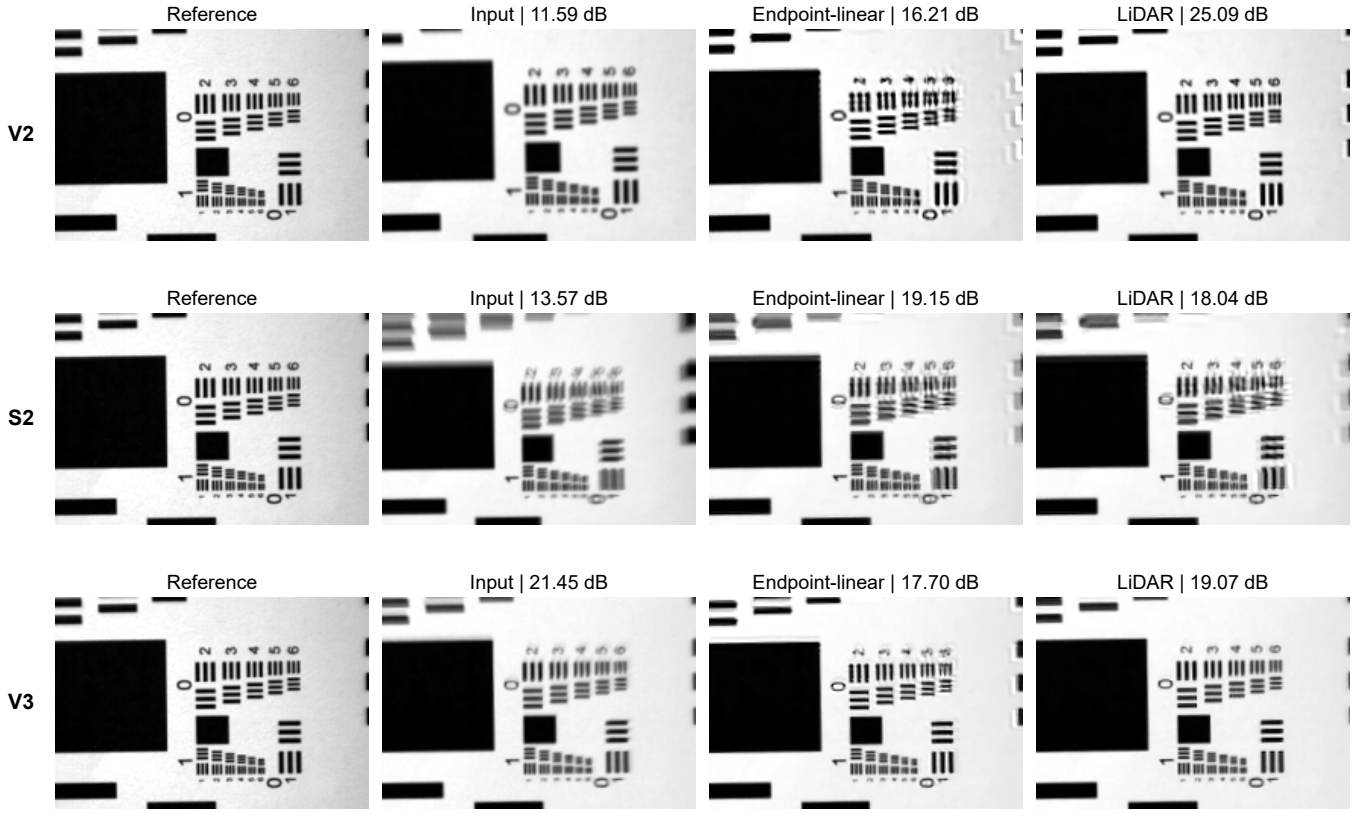


Fig. 5. Three outcomes shown with the same canonical crop and linear grayscale range $[0, 1]$. V2 has the largest measured-depth advantage; S2 favours endpoint-linear; V3 favours the input over either reconstruction. All six trials appear in Table 1 and the full analysis gallery. Scoring follows the irregular common mask within each display crop.

6.2 Examining the failed comparisons

S2 stays below endpoint-linear across all 13 one-factor settings, with paired PSNR differences from -1.209 to -0.847 dB. Changing the tested timing, TV weight, light floor, depth offset or rotation does not reverse that result. This persistence still leaves its physical cause unresolved.

For V3, denoise-only processing gives 21.43 dB, close to the 21.45 dB input, while LiDAR inversion gives 19.07 dB. Most of the loss appears when the motion model is inverted. Increasing the solver from 320 to 640 iterations leaves the score at 19.07 dB; 96 integration bins give 19.10 dB. Tested TV weights from 0.0005 to 0.02 give 19.04–19.12 dB. The loss persists in these checks. Equation (2) fits the recorded blur, so errors in the assumed motion or illumination can lead the solver away from the static reference. Pose, defocus, lighting and timing remain possible contributors; these tests do not separate their effects.

6.3 Steady-white controls and repeat variation

Table 2 summarizes all 12 primary pushes, with only three independent repeats per cell. The variable-coded cell has a larger mean model advantage than the smooth-coded cell. White-variable reconstruction beats endpoint-linear on average but loses to the input. Motion and integrated illumination differ between the white and coded pushes, preventing a causal comparison of coding.

TABLE 2
All primary cells, $n = 3$ independent pushes each. Δ_{LC} is LiDAR minus endpoint-linear PSNR (mean $\pm$ sample SD); Δ_{LI} is LiDAR minus input, also as mean $\pm$ sample SD. Per-trial scores are supplied.

Light	Instruction	Δ_{LC} (dB)	Δ_{LI} (dB)
Coded	Smooth	1.020 ± 2.694	3.174 ± 1.248
Coded	Variable	5.057 ± 3.758	3.871 ± 8.458
White	Smooth	1.549 ± 2.370	3.422 ± 2.840
White	Variable	1.396 ± 2.118	-0.932 ± 0.898

6.4 Stationary and sensitivity controls

Relative to denoise-only processing, measured-depth reconstruction gains 3.522 ± 5.411 dB on average (sample SD, $n = 6$), with five trials improving in both metrics. Thus TV denoising alone does not account for the observed mean gain.

Three static depth-driven controls change PSNR by only $+0.001$, $+0.026$ and $+0.010$ dB relative to input, with SSIM changes -0.0029, -0.0028 and -0.0022. The small changes show how the pipeline behaves without target motion; they provide a control for the moving trials.

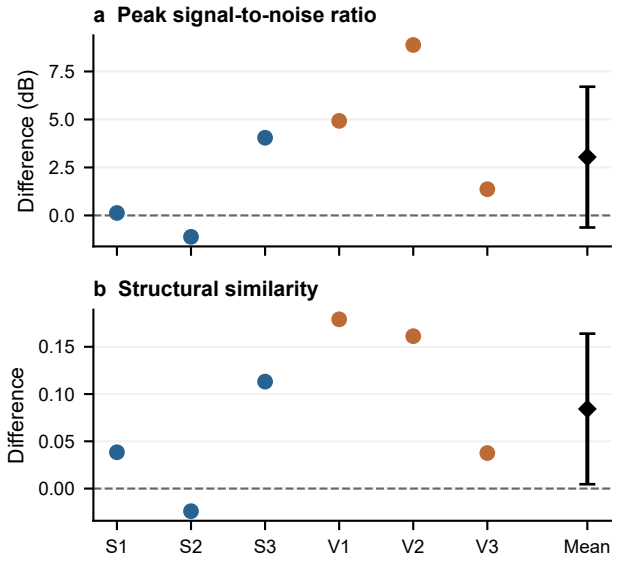


Fig. 6. LiDAR-minus-endpoint-linear differences for all six coded images. The black diamond shows the mean and one sample SD; it is separate from the six trials and does not show a confidence interval. Blue: smooth instructions; orange: variable instructions.

Thirteen one-factor checks vary TV weight, light offset, depth timing (± 50 ms), black-state floor, common depth offset (± 22 mm) and endpoint rotation. Mean coded LiDAR advantage remains 2.43–3.56 dB, while individual trial signs can change. This range describes the settings tested separately; it gives neither a statistical uncertainty bound nor a joint worst-case result. Doubling quadrature bins changes the mean advantage from 3.038 to 3.013 dB. Frozen-geometry replay in a compatible numerical environment changes PSNR by at most 0.0032 dB, showing that the frozen-geometry calculation is repeatable. Fresh registration can still change subpixel fits.

7 Limitations and Pitfalls

The illumination history is estimated from command receipts and a fitted offset. We did not measure optical transitions with a photodetector, calibrate ranging-to-receipt delay or measure rolling-row skew. The tested timing perturbations cover only selected errors.

The scale model also omits continuous tilt, lateral drift, depth-dependent defocus and spatial variation in illumination. Our fixed-focus controls show that sharpness changes even within the retained range. The optical origin remains uncertain, and the 22 mm base-to-paper offset provides no independent sensor-bias calibration. These effects were not separated experimentally, so we cannot assign either failed comparison to a single physical cause.

Reference-assisted registration and contrast normalization limit the scope of the scores. Downsampling and TV can suppress fine bars, while the shared mask still depends on reference geometry. Selection was exploratory; the archive retains every attempt and documents the weak-motion replacements. Six coded pushes and three repeats per cell support descriptive comparisons. Unmatched white/coded motion and dose prevent a causal coding claim.

8 Conclusions and Future Work

Our manual-push apparatus records long-exposure images, depth and monitor-command histories for axial motion reconstruction. Across six independent coded exposures, measured depth gives a mean PSNR gain of 3.04 dB over a matched endpoint-linear model. Five trials improve in both metrics. S2 loses to that model, while V3 loses to the blurred input. The controls and numerical checks support a conditional benefit from measured motion; timing, focus, pose and reference assistance still limit the result.

With two more months, we would first measure optical transitions, rolling-row timing, range origins, image scale and illumination along the path. We would then preregister selection rules and collect randomized repeated motion profiles at matched light dose. Continuous pose measurements and a depth-dependent defocus model would help separate possible causes of failure before testing optimized codes.

9 Reproducibility

The accompanying package contains the acquisition data, all 16 attempt scores, frozen selections and geometry, source hashes, reconstruction arrays, scripts, dependencies and build instructions. Run `python reproduce.py` to recalculate saved-array metrics and regenerate the paired-effect and mask figures. Use `--reconstruct` for all 16 RAW inversions with frozen geometry, or `--trial S1` for one coded exposure. The README covers the tested environment, hardware records and fresh registration, whose subpixel fits can vary across numerical environments. The earlier five-trial dataset is separate. Public code/data URL: *to be supplied by the authors before submission*.

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